

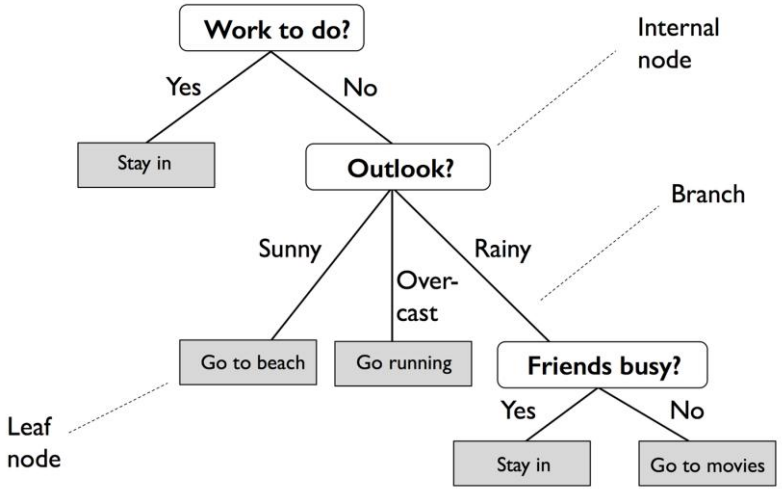
Decision Trees

COMP3314
Machine Learning

Dong Xu

Decision Tree Learning

- Decision tree classifiers are attractive models if we care about interpretability
- This model breaks down our data by making a decision based on asking a series of questions
- Example of decision tree



Decision Tree Learning

- Model learns to ask a series of questions
  - E.g.: Is sepal width ≥ 2.8 cm?
- Prediction is based on answers to questions
- What questions to ask?
  - Split data on feature that results in largest Information Gain (IG)
  - Iterative repeat splitting until leaves are pure
    - I.e., samples all belong to same class
    - This can result in a very deep tree with many nodes, which can easily lead to overfitting
    - Thus, we typically want to prune the tree by setting a limit for the maximal depth of the tree

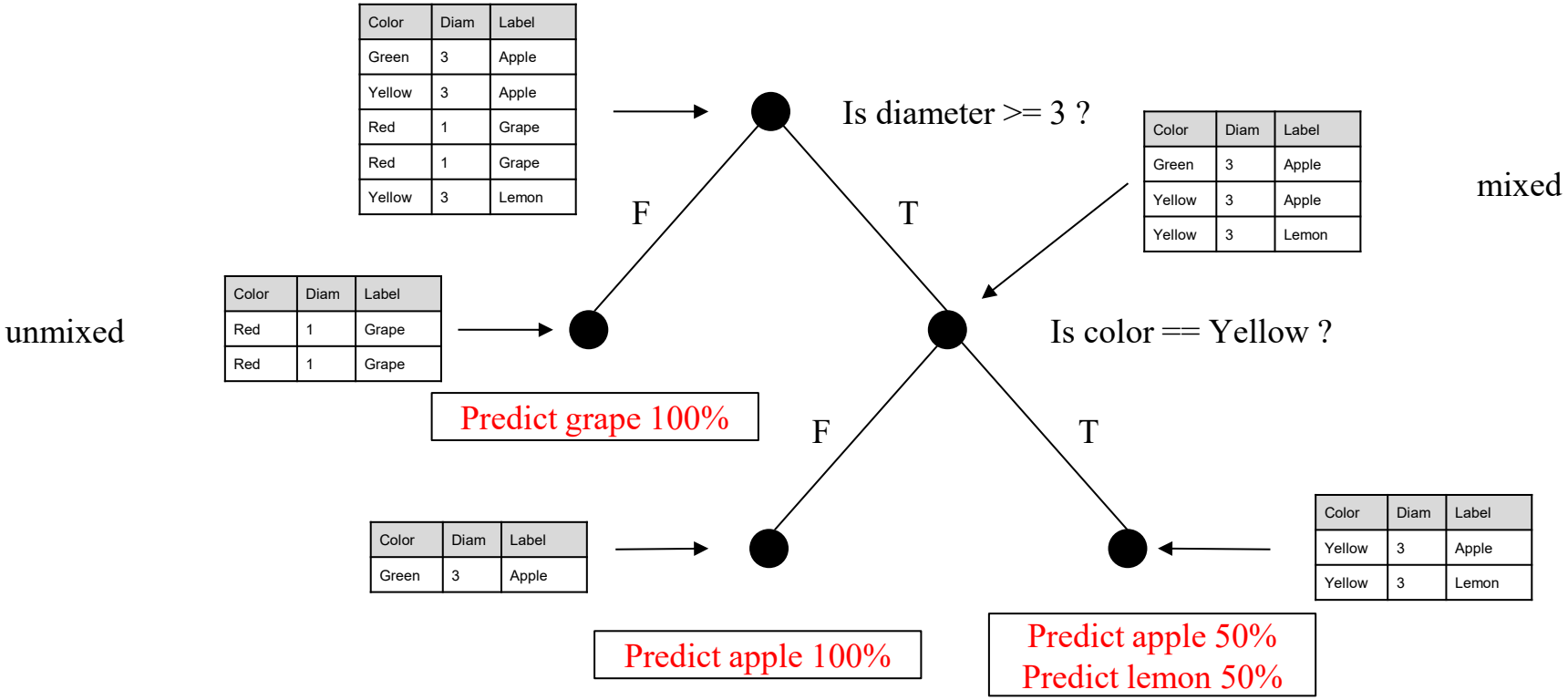
Decision Tree Learning - Intuition

- Training data:

Color	Diameter	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

features

Decision Tree Learning - Intuition



Which questions to ask and when?

- Quantify how much a question helps to unmix the labels
  - 1. Quantify the amount of uncertainty at a single node
    - E.g. Gini Impurity

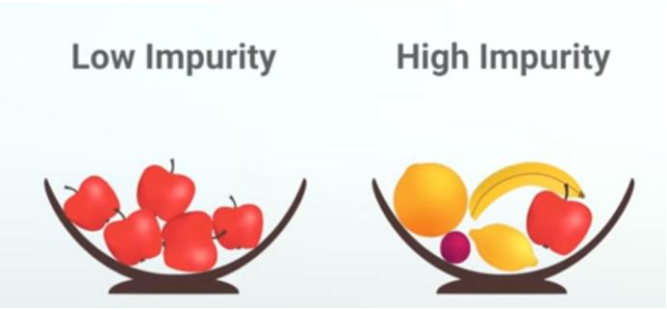
$$I_G(t)$$

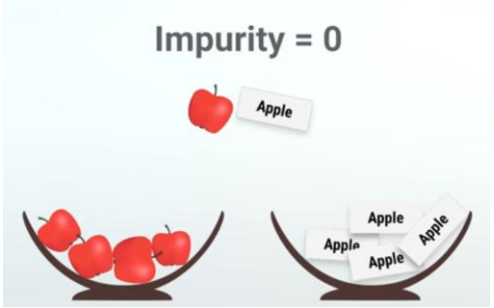
  - 2. How much does a question reduce the uncertainty
    - We aim to maximize Information Gain

$$IG(D_p, f)$$

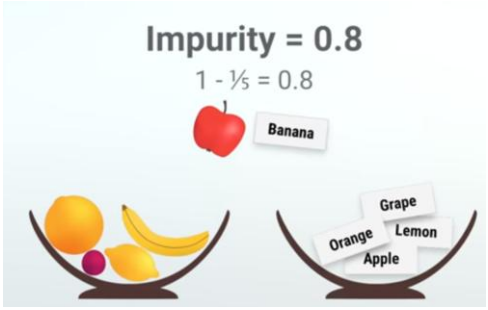
1. Gini Impurity

- Chance of being incorrect if you randomly assign a label to an example in the same set





$$I_G(t) = \sum_{i=1}^{c} p(i|t)\left(1 - p(i|t)\right) = 1 - \sum_{i=1}^{c} p(i|t)^2$$

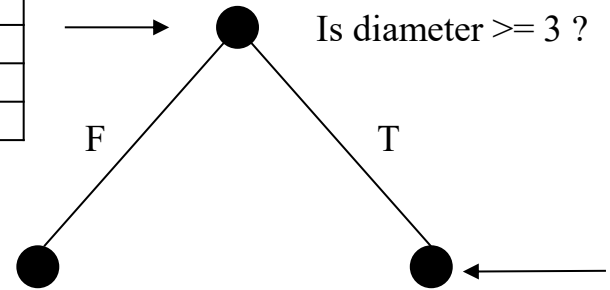


p(i | t) is the proportion of the samples that belong to class i for a particular node t

1. Gini Impurity

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon



Gini Impurity:
$I_G = 0$

Color	Diam	Label
Red	1	Grape
Red	1	Grape

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Yellow	3	Lemon

Gini Impurity:
$I_G = 0.44$

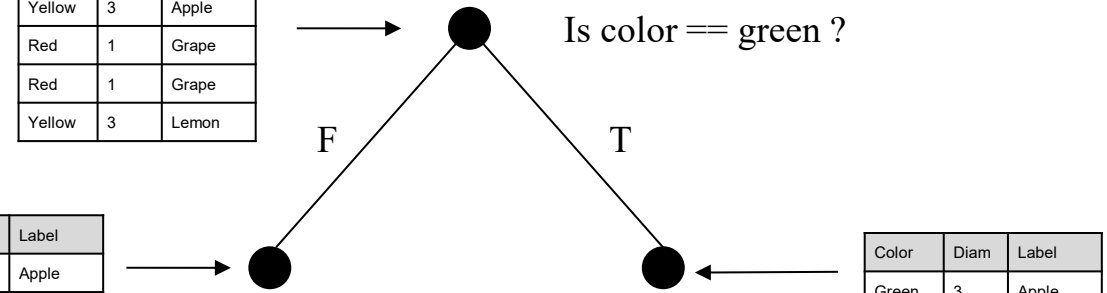
$$I_G(t) = \sum_{i=1}^{c} p(i \mid t)\left(1 - p(i \mid t)\right) = 1 - \sum_{i=1}^{c} p(i \mid t)^2$$

p(i | t) is the proportion of the samples that belong to class i for a particular node t

1. Gini Impurity

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon



Gini Impurity:
$I_G = 0.625$

Color	Diam	Label
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Color	Diam	Label
Green	3	Apple

Gini Impurity:
$I_G = 0$

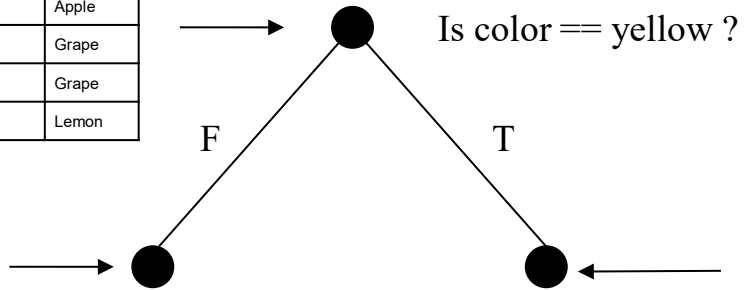
$$I_G(t) = \sum_{i=1}^{c} p(i \mid t)\left(1 - p(i \mid t)\right) = 1 - \sum_{i=1}^{c} p(i \mid t)^2$$

p(i | t) is the proportion of the samples that belong to class i for a particular node t

1. Gini Impurity

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon



Gini Impurity:
$I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Red	1	Grape
Red	1	Grape

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon

Gini Impurity:
$I_G = 0.5$

$$I_G(t) = \sum_{i=1}^{c} p(i \mid t)\left(1 - p(i \mid t)\right) = 1 - \sum_{i=1}^{c} p(i \mid t)^2$$

p(i | t) is the proportion of the samples that belong to class i for a particular node t

Which questions to ask and when?

- Quantify how much a question helps to unmix the labels
  - 1. Quantify the amount of uncertainty at a single node
    - E.g. Gini Impurity

$$I_G(t)$$

  - 2. How much does a question reduce the uncertainty
    - We aim to maximize Information Gain

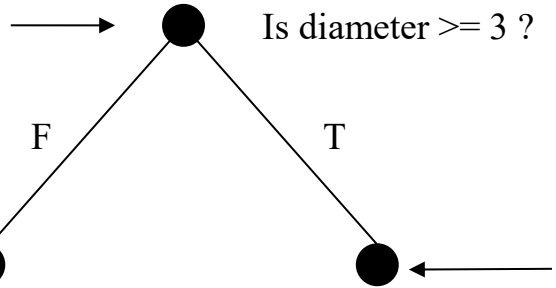
$$IG(D_p, f)$$

2. Information Gain

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Information Gain:
IG = 0.376



Gini Impurity:
$I_G = 0$

Color	Diam	Label
Red	1	Grape
Red	1	Grape

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Yellow	3	Lemon

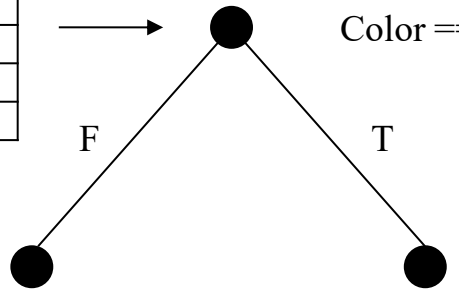
Gini Impurity:
$I_G = 0.44$

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

2. Information Gain - Task

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon



$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

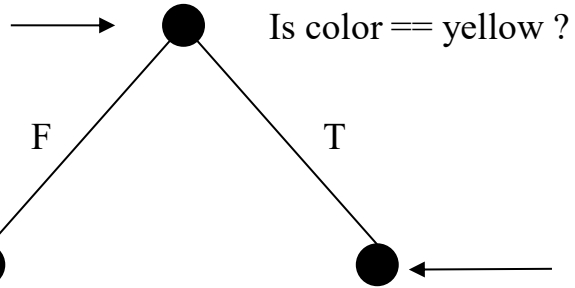
$$I_G(t) = \sum_{i=1}^{c} p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^{c} p(i|t)^2$$

2. Information Gain

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Information Gain:
IG = 0.176



Gini Impurity:
$I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Red	1	Grape
Red	1	Grape

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon

Gini Impurity:
$I_G = 0.5$

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

Example

$$I_G(t) = \sum_{i=1}^{c} p(i|t)\left(1 - p(i|t)\right) = 1 - \sum_{i=1}^{c} p(i|t)^2$$

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

Gini Impurity:
$I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Information Gain:
IG = 0.376

Is diameter >= 3 ?

F

T

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Yellow	3	Lemon

Gini Impurity:
$I_G = 0.44$

Gini Impurity:
$I_G = 0$

Color	Diam	Label
Red	1	Grape
Red	1	Grape

Is color == Yellow ?

Information Gain:
IG = 0.11

F

T

Color	Diam	Label
Green	3	Apple

Gini Impurity:
$I_G = 0$

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon

Gini Impurity:
$I_G = 0.5$

Maximizing Information Gain

- Split nodes at most informative features
  - In our tree learning algorithm we optimize an objective function
    - E.g., maximize information gain at each split

$$IG\left(D_p, f\right) = I\left(D_p\right) - \sum_{j=1}^{m} \frac{N_j}{N_p} I\left(D_j\right)$$

- f is the feature to perform the split
- D_p and D_j are the dataset of the parent and jth child node
- I is our impurity measure
- N_p is the total number of samples at the parent node, and
- N_j is the number of samples in the jth child node.

Binary Decision Tree

- For binary decision trees each parent node is split into two child nodes
  - D_{left} and D_{right}

$$IG\left(D_p, f\right) = I\left(D_p\right) - \frac{N_{left}}{N_p} I\left(D_{left}\right) - \frac{N_{right}}{N_p} I\left(D_{right}\right)$$

- The following impurity measures or splitting criteria are commonly used in binary decision trees
  - Gini impurity (I_G),
  - Entropy (I_H), and
  - Classification error (I_E)

Entropy

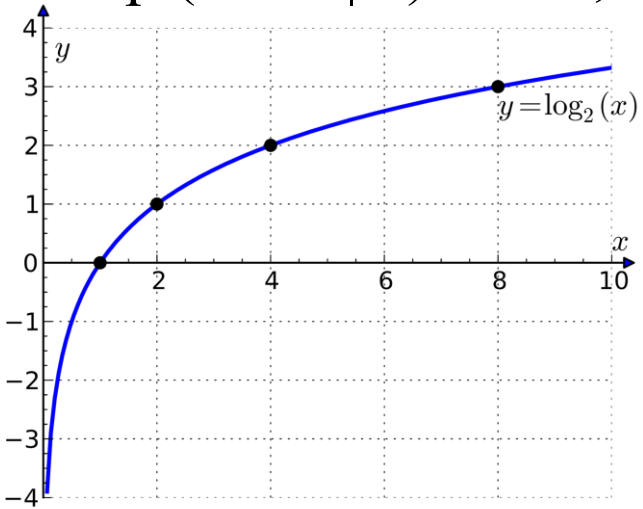
$$I_H(t) = -\sum_{i=1}^{c} p(i \mid t) \log_2 p(i \mid t)$$

- p(i | t) is the proportion of the samples that belong to class i for a particular node t
- The entropy is therefore 0 if all samples at a node belong to the same class, and the entropy is maximal if we have a uniform class distribution
- The entropy criterion attempts to maximize the mutual information in the tree

p(i | t) is the proportion of the samples that belong to class i for a particular node t

Entropy

- For example, in a binary class setting (c = 2)
  - The entropy is 0 if p (i = 1| t) = 1 or p (i = 0 | t) = 0
  - If the classes are distributed uniformly with p (i = 1| t) = 0.5 and p (i = 0 | t) = 0.5 , the entropy is 1



$$I_H(t) = -\sum_{i=1}^{c} p(i \mid t) \log_2 p(i \mid t)$$

p(i | t) is the proportion of the samples that belong to class i for a particular node t

Gini

- The Gini impurity can be understood as a criterion to minimize the probability of misclassification

$$I_G(t) = \sum_{i=1}^{c} p(i \mid t)\big(1 - p(i \mid t)\big) = 1 - \sum_{i=1}^{c} p(i \mid t)^2$$

- Similar to entropy, the Gini impurity is maximal if the classes are perfectly mixed, for example, in a binary class setting (c = 2)

$$I_G(t) = 1 - \sum_{i=1}^{c} 0.5^2 = 0.5$$

p(i | t) is the proportion of the samples that belong to class i for a particular node t

Classification Error

- Another impurity measure is the classification error

$$I_E = 1 - \max\left\{ p\left(i \mid t\right)\right\}$$

- This is a useful criterion for pruning but not recommended for growing a decision tree, since it is less sensitive to changes in the class probabilities of the nodes

p(i | t) is the proportion of the samples that belong to class i for a particular node t

Code - DecisionTrees.ipynb

- Available here on CoLab

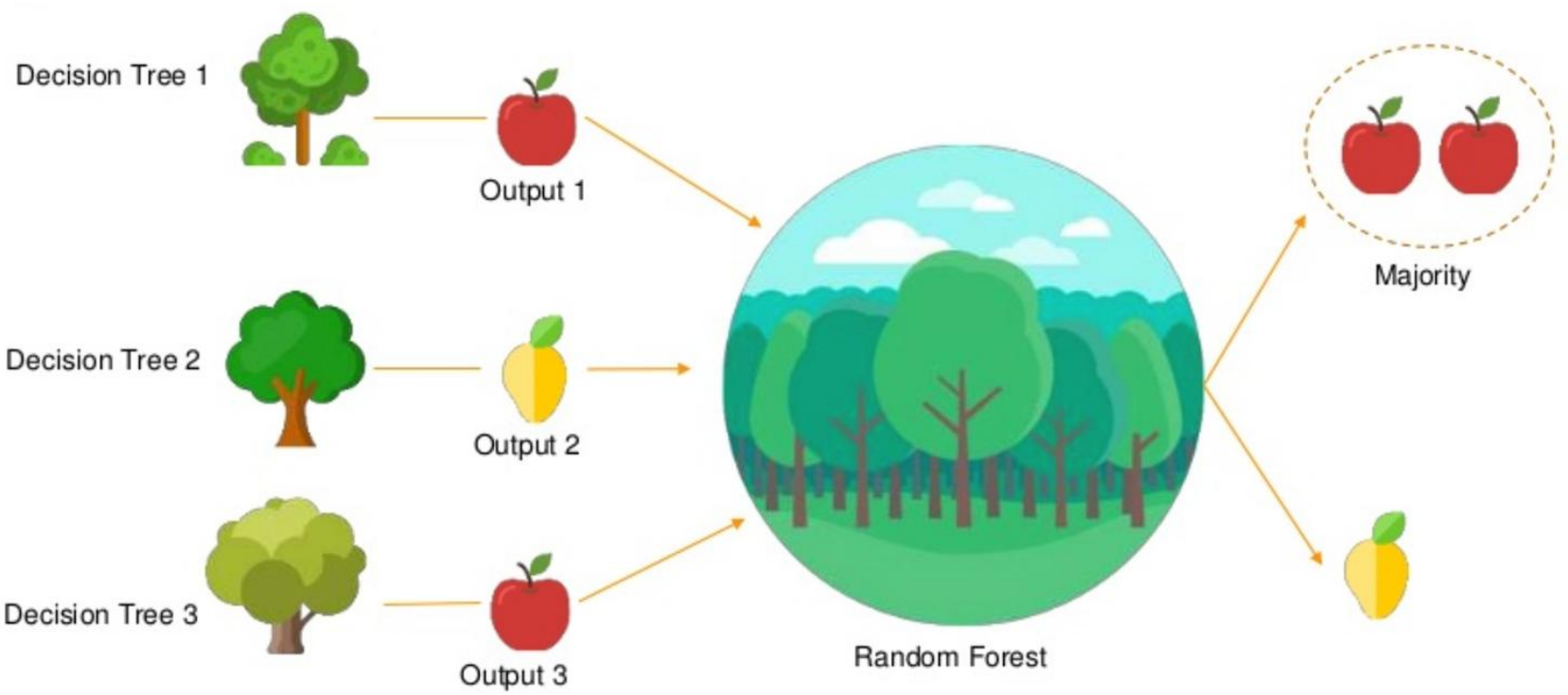
Random Forests

- A random forest can be considered as an ensemble of decision trees
- The idea behind a random forest is to average multiple (deep) decision trees that individually suffer from high variance, to build a more robust model that has a better generalization performance and is less susceptible to overfitting

Random Forests Creation

1. Draw a random bootstrap sample of size n (i.e., randomly choose n samples from the training set with replacement)
2. Grow a decision tree from the bootstrap sample. At each node:
   a. Randomly select d features without replacement
   b. Split the node using the feature that provides the best split according to the objective function, for instance, maximizing the information gain
3. Repeat steps 1. and 2. k times
4. Aggregate the prediction by each tree to assign the class label by majority vote

Random Forests Intuition



Bootstrap Sample Size

- Decreasing the size of the bootstrap samples (i.e., decreasing n) increases the diversity among the individual trees
  - The probability that a particular training sample is included in the bootstrap sample is lower
  - This increases the randomness of the random forest
    - Helps to reduce the effect of overfitting
- However, smaller bootstrap samples typically result in a lower overall performance of the random forest, a small gap between training and test performance, but a low test performance overall
- Conversely, increasing the size of the bootstrap sample may increase the degree of overfitting
  - The bootstrap samples, and consequently the individual decision trees, become more similar to each other, they learn to fit the original training dataset more closely